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J. W. Behrens and A. D. Carlson (eds.), *50 Years with Nuclear Fission*, vol. 1 & 2, Am. Nucl. Soc., La Grange Park, Illinois, 1989.
G. T. Seaborg and W. D. Loveland, *The Elements Beyond Uranium*, John Wiley & Sons Inc., New York 1990.

Journals of importance to radiochemistry and nuclear chemistry:

Radiochimica Acta, R. Oldenbourg Verlag, Munich (Vol. 99, 2011).
Journal of Radioanalytical and Nuclear Chemistry, Akadémiai Kiadó, Budapest (Vol. 290, 2011).
Applied Radiation and Isotopes, Elsevier (Vol. 69, 2011).
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Nuclear Technology, Am. Nucl. Soc., La Grange Park, Illinois (Vol. 176, 2011).

CHAPTER 2

Elementary Particles

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This chapter gives the reader an insight in the world of "sub-electron" particles. It is in no way exhaustive but is intended to tease the interest of the reader into studying more literature dedicated to this specific subject. The basics of nuclear physics is of great importance to understand the mechanism governing the physics inside a nucleus.

The electron, proton and neutron are well known elementary particles. However, high energy nuclear collisions, such as those that occur in the atmosphere when it is hit by very high energy particles from space, are known to produce a wealth of other particles which once used to be called elementary particles. Such particles can now also be formed using large particle accelerators on earth, e.g. the one in CERN.

2.1. ELEMENTARY PARTICLES

The group of elementary particles began to be considerably expanded about 1947 when physicists discovered the first of the so-called "strange" particles in cloud chamber pictures of cosmic rays, see Figure 2.1. These new elementary particles were called strange because they lived almost a million million times[1] longer than scientists had any reason to expect at that time. The population of elementary particles has literally exploded since then, as physicists have built larger and larger particle accelerators, by which it is possible to impart sufficient kinetic energy to protons so that interaction with nuclides transform a large fraction of the kinetic energy into matter. The present limit is in the multi TeV range (Ch. 16). In 2011 it is thus possible to produce particles with a mass of up to ~3500 proton masses, and hundreds of new "strange" particles have been observed. Figure 2.2 is a typical picture of a reaction observed in a liquid hydrogen bubble chamber at an accelerator center.

[1] million million times = 10^{12} times

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